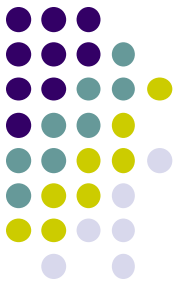


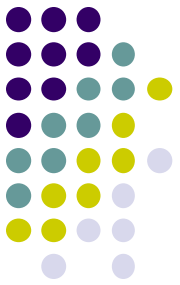
Bayesian Methods

- Our focus this lecture
- **Learning and classification** methods based on probability theory
- **Bayes theorem** plays a critical role in probabilistic learning and classification
- Uses ***prior probability*** of each category given no information about an item
- Categorization produces a ***posterior probability distribution*** over the possible categories given a description of an item



A dice-y(!) problem

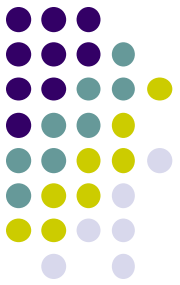
- Two dice were rolled and yielding values D_1 and D_2 .
- **A** is the event: **$D_1 + D_2 = 4$**
- $P(A) = ?$
 - $E = \{(1,3), (3,1), (2,2)\}$
 - $|S| = 36$
 - $P(E) = 3/36$
- Let **B** be the event: **$D_2 = 2$**
- **$P(A \text{ given } B \text{ already observed}) = ?$**
 - $S = \{(1,2), (2,2), (3,2), (4,2), (5,2), (6,2)\}$
 - $A = \{(2,2)\}$
 - $P(A \text{ given } B \text{ already observed}) = 1/6$



Conditional Probability

- **Conditional probability** is probability that A occurs *given that B has already occurred*
 - “Conditioning on B”
- Written as **$P(A \mid B)$**
 - Means “ **$P(A, \text{ given } B \text{ already observed})$** ”
 - Sample space, S, reduced to those elements consistent with B (i.e. $S \cap B$)
 - Event space, A, reduced to those elements consistent with B (i.e. $A \cap B$)
- **With equally likely outcomes**

$$P(A \mid B) = \frac{\text{\# of outcomes in A consistent with B}}{\text{\# of outcomes in S consistent with B}} = \frac{|A \cap B|}{|S \cap B|} = \frac{|AB|}{|B|}$$

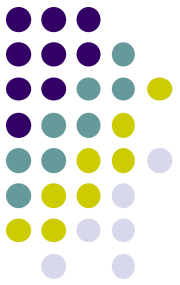


Conditional Probability

- General Definition

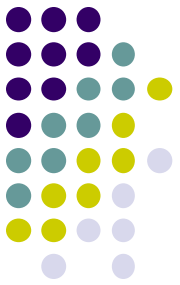
$$P(A|B) = \frac{P(AB)}{P(B)}, \text{ if } P(B) \neq 0$$

- If $P(B) = 0$, $P(A|B)$ is undefined
- Holds even when outcomes are not equally likely
- $P(AB) = P(A|B)P(B)$ [**chain or multiplication rule**]
- Similarly, $P(AB) = P(B|A)P(A)$



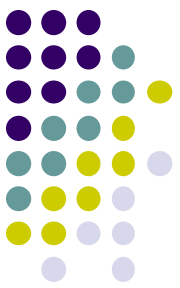
The dice-y problem-revisited

- Two dice are rolled yielding values D_1 and D_2
- Event A : $D_1 + D_2 = 4$
 - $|S| = 36$, $|A| = 3$ as, $A = \{(1,3), (2,2), (3,1)\}$
 - $P(A) = 3/36$
- Event B: $D_2 = 2$
 - $|B| = 6$ as, $B = \{(1,2), (2,2), (3,2), (4,2), (5,2), (6,2)\}$
 - $P(B) = 6/36$
- $AB = \{(2,2)\}$
 - $P(AB) = 1/36$
- $P(\text{Sum is 4, given 2 is observed on the first die})$
 - $P(A|B) = P(AB)/P(B) = 1/36 / 1/6 = 6/36 = 1/6$



Problem 3

- Suppose that two balls are to be selected at random, without replacement from a box containing
- ***r red balls and b blue balls***
- Find probability that the **first ball will be red and the second ball will be blue**
- ***Solution***
- ***A = event that the first ball is red***
 - ***$P(A) = r/(r+b)$***
- ***B = event that the 2nd ball is blue***
 - ***$P(B|A) = b/(r+b-1)$***
- ***$P(AB) = P(B|A)P(A) = rb/((r+b)(r+b-1))$***



Multiplication rule

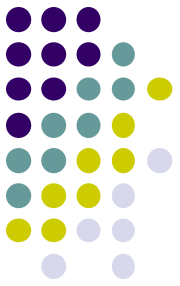
The multiplication rule

$$P(E_1 E_2 E_3 \cdots E_n) = P(E_1) P(E_2 | E_1) P(E_3 | E_1 E_2) \cdots P(E_n | E_1 \cdots E_{n-1})$$

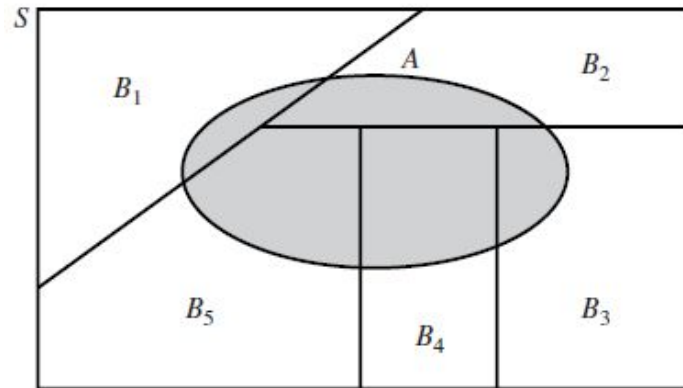
To prove the multiplication rule, just apply the definition of conditional probability to its right-hand side, giving

$$P(E_1) \frac{P(E_1 E_2)}{P(E_1)} \frac{P(E_1 E_2 E_3)}{P(E_1 E_2)} \cdots \frac{P(E_1 E_2 \cdots E_n)}{P(E_1 E_2 \cdots E_{n-1})} = P(E_1 E_2 \cdots E_n)$$

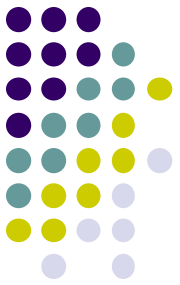
Partitions and Total Probability



- Let S = sample space of some experiment
- consider **k events $B_1, \dots, B_k$ in S** such that
 - $B_1, \dots, B_k$ are disjoint and
 - $UB_j = S, j = 1, 2, \dots, k$
- Then these events form **a partition of S**



- Here, $B_1, B_2, B_3, B_4,$ and B_5 form a partition of S



Total Probability

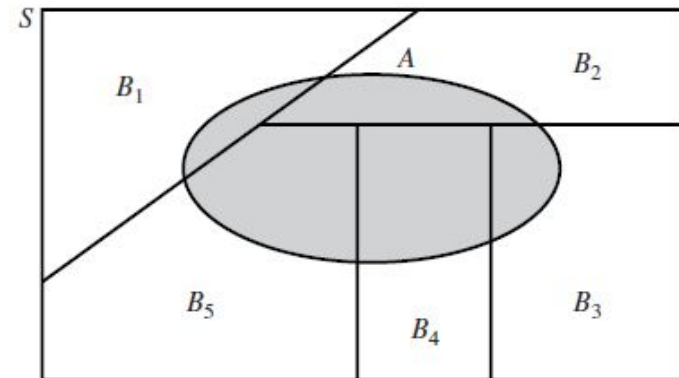
- Suppose that the events $B_1, \dots, B_k$ form a partition of the space S and
- $\Pr(B_j) > 0$, for $j = 1, \dots, k$.
- for every event A in S :

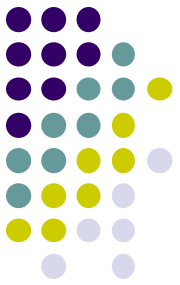
$$P(A) = \sum_{j=1}^k P(B_j)P(A | B_j)$$

- **Proof:**
- $A \cap B_1, A \cap B_1, \dots, A \cap B_k$ forms a partition of A
- $A = (A \cap B_1) \cup \dots \cup (A \cap B_k)$
- As $(A \cap B_j)$ s are **mutually disjoint**

$$P(A) = \sum_{j=1}^k P(A \cap B_j)$$

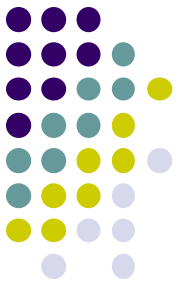
$$= \sum_{j=1}^k P(B_j)P(A | B_j), \text{ since } P(B_j) \neq 0$$





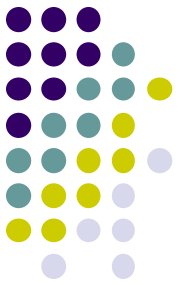
Problem 1

- Box 1 contains 60 red balls and 40 white balls
- Box 2 contains 10 red balls and 20 white balls
- **One box is selected at random and a ball is selected at random from that box**
- Let event **A = red ball selected**, $P(A) = ?$
- **Solution:**
- **B_1 = box 1 is selected**
- **B_2 = box 2 is selected**
- **$P(A) = P(B_1)P(A|B_1) + P(B_2)P(A|B_2)$
 $= \frac{1}{2} * \frac{60}{100} + \frac{1}{2} * \frac{10}{30} = \frac{7}{15}$**



Problem 2

- Same experiment as problem 1
- Now **after selection we observe that the ball selected is Red**
- What is the probability that it came from **box 1**?
 - $P(B_1|A)$
- What is the probability that it came from **box 2**?
 - $P(B_2|A)$
- **Solution:**
- $$P(B_1|A) = P(B_1A)/P(A) = P(B_1)P(A|B_1)/P(A)$$
$$= (1/2 * 60/100)/(7/15)$$
$$= 9/14$$
- Similarly, $P(B_2|A) = P(B_2)P(A|B_2)/P(A)$



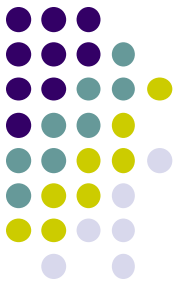
Bayes' Theorem

- Suppose that the events $B_1, \dots, B_k$ form a **partition of the space S** and **$\Pr(B_j) > 0$** , for $j = 1, \dots, k$.
- Let A be an event in S such that **$P(A) > 0$**
- then for $i = 1, \dots, k$

$$P(B_i | A) = \frac{P(B_i)P(A | B_i)}{\sum_{j=1}^k P(B_j)P(A | B_j)}$$

- **Proof:**

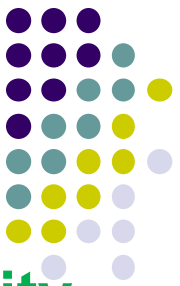
$$P(B_i | A) = \frac{P(B_i A)}{P(A)} = \frac{P(B_i)P(A | B_i)}{\sum_{j=1}^k P(B_j)P(A | B_j)}$$



Problem 3

- M_1 produces 20% of the items
- M_2 produces 30% of the items
- M_3 produces 50% of the items
- 1% of the items produced by machine M_1 are defective
- 2% of the items produced by machine M_2 are defective
- 3% of the items produced by machine M_3 are defective.
- one item is selected at random from the entire batch and it is found to be defective
- determine the probability that this item was produced by machine M_2
- **Solution:**
- Let B_i = item selected is produced by machine M_i
- $P(B_1) = .2$, $P(B_2) = .3$, and $P(B_3) = .5$
- Let A = selected item is defective
- $P(A|B_1) = .01$, $P(A|B_2) = .02$, and $P(A|B_3) = .03$

$$P(B_2|A) = \frac{P(B_2)P(A|B_2)}{\sum_{j=1}^3 P(B_j)P(A|B_j)} = \frac{(.3)(.02)}{(.2)(.01) + (.3)(.02) + (.5)(.03)} = .26$$



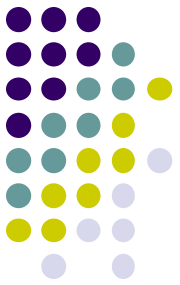
Bayes' Theorem

Posterior Probability **Likelihood** **Prior Probability**

$$P(H | E) = \frac{P(E | H)P(H)}{P(E)}$$

- What is $P(E)$?
 - $P(E) = P(E|H)P(H) + P(E|H^c)P(H^c)$
 - Why?
- We need $P(E|H)$, $P(E|H^c)$
 - In this problem, $P(E|H) = 1$
 - $P(E|H^c) = .2$
- So, our posterior belief that the defendant is guilty is

$$P(H | E) = \frac{1(.6)}{1(.6) + (.2)(1-.6)} \approx 0.882$$



Problem 4

- A test is **98% effective** at detecting a disease
- Test has a “**false positive**” rate of 1%
- 0.5% of the population has this disease
- Let E = you **test positive** for the disease with this test
- Let H = you **actually have the disease**
- $P(H | E)$?

- **Solution:**

- $P(E|H) = .98$

- $P(E|H^c) = .001$

- $P(H) = .005$

$$\begin{aligned} P(H | E) &= \frac{P(E | H)P(H)}{P(E | H)P(H) + P(E | H^c)P(H^c)} \\ &= \frac{(0.98)(0.005)}{(0.98)(0.005) + (0.01)(1 - 0.005)} \approx .33 \end{aligned}$$

Bayes' Theorem

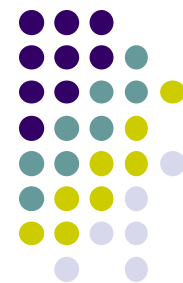


- Given a **hypothesis h** and **data D** which bears on the hypothesis:

$$P(h | D) = \frac{P(D | h)P(h)}{P(D)}$$

- $P(h)$: independent probability of hypothesis h : **prior probability of hypothesis**
- $P(D)$: independent probability of data D
- $P(D|h)$: conditional probability of data D , given hypothesis h is true. (called **likelihood** of data D , given hypothesis h)
- $P(h|D)$: conditional probability of hypothesis h being true, given data D : **posterior probability of hypothesis**
- Question:** Write the Bayes' theorem on Posterior Probability and identify the following terms — prior and posterior probability, likelihood, independent probability.

Maximum A Posterior



Question: Explain how to compute the Maximum A Posterior (MAP) hypothesis h_{MAP} for some given data D .

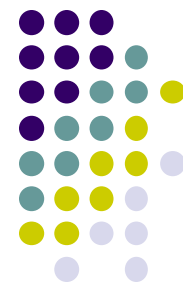
- Based on Bayes' Theorem, we can compute the **Maximum A Posterior (MAP) hypothesis h** for the given data D
- We are interested in the best hypothesis for some space H given observed training data D .

$$\begin{aligned} h_{MAP} &\equiv \operatorname{argmax}_{h \in H} P(h \mid D) \\ &= \operatorname{argmax}_{h \in H} \frac{P(D \mid h)P(h)}{P(D)} \\ &= \operatorname{argmax}_{h \in H} P(D \mid h)P(h) \end{aligned}$$

H : set of all hypothesis.

Note that we can drop $P(D)$ as the probability of the data is constant (and independent of the hypothesis).

Maximum Likelihood



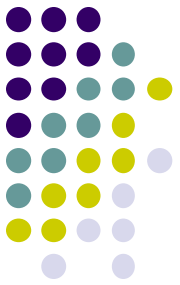
Question: Explain how to compute the Maximum Likelihood hypothesis h_{ML} for some given data D . What is the assumption for Maximum Likelihood hypothesis?

- Now assume that all hypotheses are **equally probable a priori**,
 - i.e., $P(h_i) = P(h_j)$ for all h_i, h_j belong to H .
- This is called assuming a **uniform prior**. It simplifies computing the posterior:

$$h_{ML} = \arg \max_{h \in H} P(D | h)$$

- This hypothesis is called the **maximum likelihood hypothesis**. (because, $P(D|h)$ is called the likelihood of Data D , given the hypothesis h is true)

Desirable Properties of Bayes Classifier



Question: What are the desirable properties of the Bayes' Classifier? Or, Write three desirable properties of the Bayes' classifier.

- ***Incrementality:*** with each training example, the prior probability and the data likelihood can be updated dynamically. This is flexible and robust to errors.
- ***Combines prior knowledge and observed data:*** Conceptually Sound and beautiful! The prior probability of a hypothesis multiplied with probability of the data, given the hypothesis
- ***Probabilistic hypothesis:*** Outputs not only a classification, but a probability distribution over all classes

= $P(C) \cdot P(D|C) / P(D)$ [$P(D|C)$ = very easy, but $P(C|D)$ = difficult]
 (likelihood) (posterior probab.)

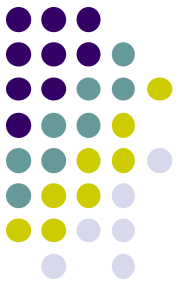
Suppose Two classes c1 and c2. Then, test data x observed. To which class should you classify (assign) the data x ?

Find posterior probabilities: $P(c1|x)$ and $P(c2|x)$ in above way

$$P(c1|x) = P(c1) * P(x|c1) / P(x) ; \quad P(c2|x) = P(c2) * P(x|c2) / P(x)$$

Assign x to the class with **Higher** posterior probability value

* Same Denominator $P(x)$, so you can compute **Numerator only**



- Now see the Salmon and Bass fish example (03-MLE Classifier.pdf)

Example Data: 'Play Cricket' data



Day	Outlook	Temperature	Humidity	Wind	Play Cricket
Day1	Sunny	Hot	High	Weak	No
Day2	Sunny	Hot	High	Strong	No
Day3	Overcast	Hot	High	Weak	Yes
Day4	Rain	Mild	High	Weak	Yes
Day5	Rain	Cool	Normal	Weak	Yes
Day6	Rain	Cool	Normal	Strong	No
Day7	Overcast	Cool	Normal	Strong	Yes
Day8	Sunny	Mild	High	Weak	No
Day9	Sunny	Cool	Normal	Weak	Yes
Day10	Rain	Mild	Normal	Weak	Yes
Day11	Sunny	Mild	Normal	Strong	Yes
Day12	Overcast	Mild	High	Strong	Yes
Day13	Overcast	Hot	Normal	Weak	Yes
Day14	Rain	Mild	High	Strong	No

Attributes

Class

***Question:** For the day <sunny, hot, normal, strong>, what's the play prediction?

Notations

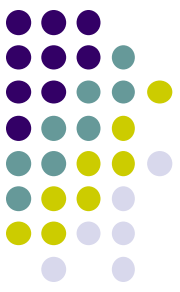


- D is the data table given
 - Each row is an example
- We have **4 attributes**
 - $x_1 = \text{outlook}$, possible values = {sunny, overcast, rain}
 - $x_2 = \text{temperature}$, possible values = {hot, mild, cool}
 - $x_3 = \text{humidity}$, possible values = {high, normal}
 - $x_4 = \text{wind}$, possible values = {weak, strong}
- Classes $C = \{\text{yes, no}\} = \{\mathbf{Y}, \mathbf{N}\}$
- In general,
 - We are given **training data** D
 - D contains instances of different classes $c_j \in C$
 - Each instance $\mathbf{x}$ contains the values for the attributes $\mathbf{x} = \langle x_1, x_2, \dots, x_n \rangle$

Bayesian Classification: Motivation



- Classification can be written as:
 - **Given an object x**
 - **with attribute $x_1, \dots, x_n$ having values $v_1, \dots, v_n$**
 - **what is the probability that object x belong to class c_j ?**
 - Given day is $\langle \text{sunny, hot, high, strong} \rangle$, what's the probability of playing cricket?
 - **$P(x \in Y \mid \text{outlook} = \text{sunny, temperature} = \text{hot, humidity} = \text{high, wind} = \text{strong})$**
 - **In probability notation: $P(x \in c_j \mid x_1=v_1, \dots, x_n=v_n)$?**
 - Use simplified notation: **$P(c_j \mid X)$**
 - **Informally, Find $P(Y \mid \langle \text{sunny, hot, normal, strong} \rangle)$ and $P(N \mid \langle \text{sunny, hot, normal, strong} \rangle)$**



Bayes Classifiers: Key idea:

Assign the **most probable class** c_{MAP} using Bayes' theorem

$$\begin{aligned} c_{MAP} &= \operatorname{argmax}_{c_j \in C} P(c_j | X) \\ &= \operatorname{argmax}_{c_j \in C} \frac{P(X | c_j)P(c_j)}{P(X)} \\ &= \operatorname{argmax}_{c_j \in C} P(X | c_j)P(c_j) \end{aligned}$$

$$= \operatorname{argmax}_{c_j \in C} P(x_1 = v_1, x_2 = v_2, \dots, x_n = v_n | c_j)P(c_j)$$

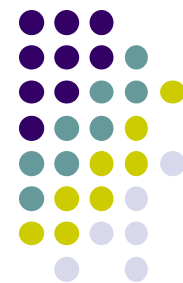
means : $P(x_1=\text{sunny}, x_2=\text{hot}, x_3=\text{Normal}, x_4=\text{Strong} | Y) * P(Y)$

and : $P(x_1=\text{sunny}, x_2=\text{hot}, x_3=\text{Normal}, x_4=\text{Strong} | N) * P(N)$

called naive Bayes, because estimates these complex probabilities as

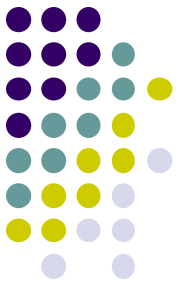
$$P(x_1=\text{sunny}|Y) * P(x_2=\text{hot}|Y) * P(x_3=\text{Normal}|Y) * P(x_4=\text{Strong} | Y) * P(Y)$$

Bayesian Classification: Motivation



$$\operatorname{argmax}_{c_j \in C} P(X | c_j) P(c_j)$$

- What does each term mean?
 - **$P(X | c_j) = P(x_1=v_1, \dots, x_n=v_n | x \in c_j)$**
 - Probability that the attributes $x_1, \dots, x_n$ will have values $v_1, \dots, v_n$ given an object x is in class c_j
 - **$P(c_j)$: Prior probability of class c_j**
 - Probability that a random object selected is of class c_j
 - Note: **$P(X) = P(x_1=v_1, \dots, x_n=v_n)$**
 - Probability that a random object selected will have values $v_1, \dots, v_n$ for attributes $x_1, \dots, x_n$

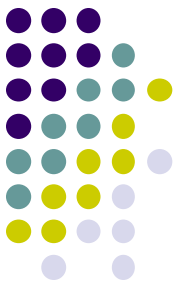


Estimation

- But how to estimate the probabilities?
 - Draw a sample!
 - Which, in classification term, means look at the training set

Parameters Estimation

Question: What is assumption of Naïve Bayes classifier (NBC)? Write and explain the Independence assumption of NBC. Why is it called the Naïve Bayes classifier?

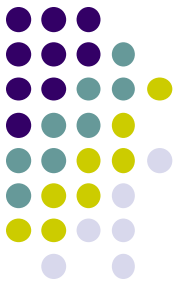


- $P(c_j)$ Can be estimated from the **frequency of classes** in the training data D. **Ans (last Part):** Because of the Independence Assumption, which is usually not True. For example, the probability of humidity=high and temp = cool is not independent of (actually, highly Correlated to) the Outlook = “rainy”
- $P(x_1, x_2, \dots, x_n | c_j)$
 - $O(|X|^n \cdot |C|)$ possible combinations of the parameters
 - Could only be estimated if a very, very large number of training examples was available.
- **Independence Assumption:** attribute values are conditionally **independent** of each other given the target value: this is why it is called **naïve Bayes' Classifier**.

$$P(x_1, x_2, \dots, x_n | c_j) = \prod_i P(x_i | c_j)$$

$$c_{NB} = \arg \max_{c_j \in C} P(c_j) \prod_i P(x_i | c_j)$$

Properties



- Estimating $P(x_i | c_j)$ instead of $P(x_1, x_2, \dots, x_n | c_j)$ **greatly reduces the number of parameters** (and the data sparseness).
- The learning step in Naïve Bayes' consists of estimating $P(x_i | c_j)$ and $P(c_j)$ based on **the frequencies** in the training data
- An unseen instance is classified by computing the class that **maximizes the posterior**
- When **conditioned independence is satisfied**, Naïve Bayes corresponds to **MAP classification**.

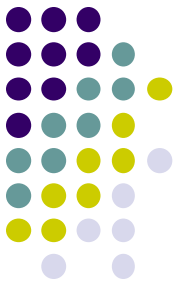
Example. 'Play Cricket' data



Day	Outlook	Temperature	Humidity	Wind	Play Cricket
Day1	Sunny	Hot	High	Weak	No
Day2	Sunny	Hot	High	Strong	No
Day3	Overcast	Hot	High	Weak	Yes
Day4	Rain	Mild	High	Weak	Yes
Day5	Rain	Cool	Normal	Weak	Yes
Day6	Rain	Cool	Normal	Strong	No
Day7	Overcast	Cool	Normal	Strong	Yes
Day8	Sunny	Mild	High	Weak	No
Day9	Sunny	Cool	Normal	Weak	Yes
Day10	Rain	Mild	Normal	Weak	Yes
Day11	Sunny	Mild	Normal	Strong	Yes
Day12	Overcast	Mild	High	Strong	Yes
Day13	Overcast	Hot	Normal	Weak	Yes
Day14	Rain	Mild	High	Strong	No

***Question:** For the day <sunny, hot, normal, strong>, what's the Play prediction?

Naive Bayes' Classifier – Computations



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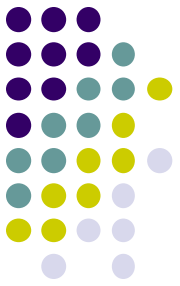
- Given a D, we can compute the probabilities $P(x_i | c_j)$

Outlook	Y	N		Humidity	Y	N
sunny	2/9	3/5		high	3/9	4/5
overcast	4/9	0		normal	6/9	1/5
rain	3/9	2/5				
Temperature				Windy		
hot	2/9	2/5		strong	3/9	3/5
mild	4/9	2/5		weak	6/9	2/5
cool	3/9	1/5				

- $P(\text{sunny} | C_Y) = 2/9$
- $P(\text{sunny} | C_N) = 3/5$

- $P(\text{normal} | C_Y) = 6/9$
- $P(\text{strong} | C_N) = 3/5$

Naive Bayes' Classifier – Computations



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- Now with object $x = (\text{sunny}, \text{hot}, \text{normal}, \text{strong})$

$$P(C_Y) P(X|C_Y) = 9/14 * 2/9 * 2/9 * 6/9 * 3/9 = 0.007$$

$$P(C_N) P(X|C_N) = 5/14 * 3/5 * 2/5 * 1/5 * 3/5 = 0.01$$

$$P(C_Y) = 9 / 14$$

$$P(C_N) = 5 / 14$$

x is in class C_N ('NO' Play Cricket)

Underflow Prevention



Question: Describe the source of floating point underflow problem in the Naïve Bayes Classifier. Explain how this problem can be solved.

- Multiplying lots of probabilities, which are between 0 and 1 by definition, can result in floating-point underflow. Also called Vanishing Zero problem
- Since $\log(xy) = \log(x) + \log(y)$,
 - it is better to perform all computations by summing logs of probabilities rather than multiplying probabilities.
- Class with highest final log sum of probabilities is still the most probable. (underflow prevention by Log)

$$c_{UPL} = \operatorname{argmax}_{c_j \in C} \log P(c_j) + \sum_{i \in \text{positions}} \log P(x_i | c_j)$$

Naïve Bayes' Classifier – Parameter Estimation



Question: What is the problem associated with Continuous attribute values in a Naïve Bayes classifier? How this problem is solved and what is the underlying assumption?

- More detail in estimating probabilities:
- If i-th attribute is **categorical**:
 $P(x_i|c_j)$ is estimated as the relative freq of samples having value x_i as i-th attribute in class c_j
- If i-th attribute is **continuous**:
 $P(x_i|c_j)$ can be estimated through a **Gaussian density function on the attribute's values**. This called 'Gaussian Distribution Assumption' of Naïve Bayes Classifier, which assumes that **each continuous attribute is distributed by Gaussian (μ, σ^2) distribution**. The mean μ and variance σ^2 can be estimated by observing the Trainig data Table D
- Computationally easy in both cases

Estimation for Continuous attributes –

Q: For the following Table, classify the Test case X: <No, Single, 120K> using Naïve Bayes Classifier

Tid	Refund	Marital Status	Taxable Income	Evade
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

$$\begin{array}{l} \text{Mean} \\ \mu = \bar{x} = \frac{\sum x}{n} \end{array} \quad \begin{array}{l} \text{Variance} \\ \sigma^2 = \frac{\sum (x - \bar{x})^2}{(n - 1)} \end{array}$$

- Normal distribution:

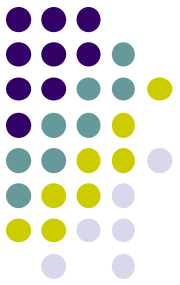
$$P(x_i | c_j) = \frac{1}{\sqrt{2\pi\sigma_j^2}} e^{-\frac{(x_i - \mu_j)^2}{2\sigma_j^2}}$$

- One for each class' each continuous attribute
- For (Income, Class=No):
 - If Class=No
 - sample mean = 110
 - sample variance = 2975

$$P(\text{Income} = 120 | \text{No}) = \frac{1}{\sqrt{2\pi(54.54)}} e^{-\frac{(120-110)^2}{2(2975)}} = 0.0072$$

Estimation for Continuous attributes –

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- Normal distribution:

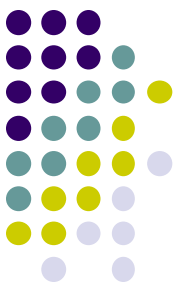
$$P(x_i | c_j) = \frac{1}{\sqrt{2\pi\sigma_j^2}} e^{-\frac{(x_i - \mu_j)^2}{2\sigma_j^2}}$$

- One for each Continuous attribute each class
- For (Income, Class=Yes):
 - If Class=Yes
 - sample mean = 90
 - sample variance = 25

$$P(\text{Income} = 120 | \text{Yes}) = \frac{1}{\sqrt{2\pi(7.07)}} e^{-\frac{(120-90)^2}{2(50)}} = 0.00000696$$

Estimation for Continuous attributes –

Q: For the following Table, classify the Test case X: <No, Single, 120K> using Naïve Bayes Classifier



Tid	Refund	Marital Status	Taxable Income	Evade
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3	No	Single	70K	No
4	Yes	Married	120K	No
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6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

- Test case **X**: <No, Single, 120K>

$$P(C_Y) = 3/10 \quad P(C_N) = 7/10$$

$$\begin{aligned} P(C_Y) P(X|C_Y) &= \\ &= 3/10 * 3/3 * 2/3 * 0.00000696 \\ &= 0.000001392 \end{aligned}$$

$$\begin{aligned} P(C_N) P(X|C_N) &= \\ &= 7/10 * 4/7 * 2/7 * 0.0072 = \\ &0.000823 \end{aligned}$$

x is in class C_N (Evade = 'NO' class)

Understanding Naive Bayes with an Example

Let's understand naive Bayes with one more example—to predict the weather based on three predictors: humidity, temperature and wind speed. The training data is the following:

Humidity	Temperature	Wind Speed	Weather
Humid	Hot	Fast	Sunny
Humid	Hot	Fast	Sunny
Humid	Hot	Slow	Sunny
Not Humid	Cold	Fast	Sunny
Not Humid	Hot	Slow	Rainy
Not Humid	Cold	Fast	Rainy
Humid	Hot	Slow	Rainy
Humid	Cold	Slow	Rainy

We'll use naive Bayes to predict the weather for the following test observation:

Question:

Humidity %	Temperature (C)	Wind Speed (Km/h)	Weather
Humid	Cold	Fast	?

We have to determine which posterior is greater, sunny or rainy.

Question: What are the Applications of Naïve Bayes Classifier?



Naive Bayes classifiers can be trained very efficiently in a supervised learning setting. In many practical applications, parameter estimation for naive Bayes models uses the method of maximum likelihood. Despite their naive design and apparently oversimplified assumptions, naive Bayes classifiers have worked quite well in many complex real-world situations.

- **Recommendation System:** Naive Bayes classifiers are used in various inferencing systems for making certain recommendations to users out of a list of possible options.
- **Real-Time Prediction:** Naive Bayes is a fast algorithm, which makes it an ideal fit for making predictions in real time.
- **Multiclass Prediction:** This algorithm is also well-known for its multiclass prediction feature. Here, we can predict the probability of multiple classes of the target variable.
- **Sentiment Analysis:** Naive Bayes is used in sentiment analysis on social networking datasets like Twitter* and Facebook* to identify positive and negative customer sentiments.
- **Text Classification:** Naive Bayes classifiers are frequently used in text classification and provide a high success rate, as compared to other algorithms.
- **Spam Filtering:** Naive Bayes is widely used in spam filtering for identifying spam email.